# Brennen A. Hill

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## Education

**University of Southern California (USC)**; Los Angeles, CA — *Anticipated May 2031*
*Ph.D. in Computer Engineering*
- **Advisor:** Dr. Priyadarshini (Priya) Panda, Intelligent Computing Lab.
- **Research Focus:** Neuroscience-inspired machine learning.
- **Honors:** Annenberg Fellowship, Viterbi School of Engineering & Graduate School Fellowship.

**University of Wisconsin-Madison**; Madison, WI — *May 2026*
*Bachelor of Science in Computer Science* — cs.wisc.edu/2026/04/08/brennen-hill-x26-is-building-ai-inspired-by-the-brain
- **GPA:** 3.8/4.0.
- **Honors:** Letters & Science Dean's List, Honors in Computer Science Program, Letters & Science Honors Program, Computer Science Department Interview & Feature.
- **Graduate AI Coursework:** Foundation Models, Learning-Based Image Synthesis, Advanced Robotics, Advanced Reinforcement Learning, Advanced Seminar (audit), Theory of Multi-agent ML.
- **AI Coursework:** Directed Study on LLMs with Dr. Sala, Game AI, Artificial Neural Networks, Artificial Intelligence, Game Theory & Learning.
- **Neuro Coursework:** Molecular & Cellular Mechanisms of Memory, Neurobiology, Computational Neuroscience.

**National University of Singapore**; Singapore — *May 2025*
*Exchange Scholar*
- Awarded a merit-based placement through a highly competitive, university-wide application process.
- **GPA:** 3.8/4.0 (calculated by UW-Madison).
- **Graduate AI Coursework:** Neural Networks & Deep Learning 2 (audit).
- **Graduate Neuro Coursework:** Frontiers in Neurotechnology (audit), Behavioral & Cognitive Neuroscience (audit).
- **AI Coursework:** AI Planning & Decision Making, Mind & Machine (both).
- **Neuro/Bio Engineering Coursework:** Mind & Machine, Organoid Engineering (audit), Bioinformatics.

## Research Experience

**Dr. Priya Panda's Intelligent Computing Lab**; University of Southern California — *Aug 2026 - Present*
*Doctoral Researcher*
- Researching the applications of the brain's learning mechanisms to address challenges in machine learning.

**Wisconsin Neuromorphic Computing and NeuroAI Lab**; University of Wisconsin-Madison — *Jun 2024 - Present*
*Founding Director & Research Lead* — neuromorphic.cs.wisc.edu/people
- Founded and direct to explore the intersection of neuroscience and artificial intelligence.
- Secured formal funding, dedicated space, support from Dr. Akhilesh Jaiswal as advisor, partnership with NeuroAI startup FinalSpark, and administrative support from UW-Madison.
- Lectured on neuroAI to audiences of over 100 and organized weekly workshops.
- Mentored 15 researchers from initial project proposals to research papers and advised 100.
- *Publications from this research:* [2], [5], [3].

**Dr. Saket Navlakha's Lab at Cold Spring Harbor Laboratory**; Long Island, NY — *May 2026 - Jul 2026*
*NeuroAI Research Intern*
- Researched and developed algorithms to enable AI to smell based on olfaction in the brain.
- Prepared for deployment onto hardware in hospitals for reducing employee workload.

**Dr. Frederic Sala's Sprocket Lab**; University of Wisconsin-Madison — *Sep 2024 - May 2026*
*Foundation Models Researcher*
- Researched novel methods for editing representations within LLMs including coarse-to-fine pipelines.
- Worked with big data, GPU clusters, and high throughput systems.
- *Publications from this research:* [7], [8].

**Dr. Josiah Hanna's Badger RL Lab**; University of Wisconsin-Madison — *May 2024 - May 2025*
*Reinforcement Learning Researcher*
- Designed reinforcement learning policies to train robots for autonomous soccer, focusing on multi-agent coordination.
- Won in RoboCup International Robotics Competition for multiple years by applying reinforcement learning research.
- *Publications from this research:* [1] (in collaboration with Dr. Berland's Lab), [6].

**Dr. Matthew Berland's Complex Play Lab**; University of Wisconsin-Madison — *Apr 2024 - May 2026*
*Reinforcement Learning Researcher*

- Architected world models and inter-agent communication strategies in multi-agent reinforcement learning (MARL).
- Developed a novel adversarial co-evolution framework to automatically generate curriculum for MARL.
- *Publications from this research:* [1] (in collaboration with Dr. Hanna's Lab), [4].

**Collaboration with Dr. Rajendra Kumar-Singh**; Tufts University (Remote) *Oct 2025 - Aug 2026*
*NeuroAI Research Collaborator & AI Lead*
- Personally recruited by Dr. Kumar-Singh to serve as the AI lead for two proposed biomedical ventures.
- Authored roadmaps analyzing competitive landscape, resource requirements, and strategy for (1) genAI platform for de novo AAV gene-therapy vector design & (2) clinically-tailored AI vision aid for patients with AMD & Glaucoma.

**Mentorship from Dr. Michael Levin**; Allen Discovery Center at Tufts University (Remote) *Oct 2025 - Aug 2026*
*NeuroAI Research Mentee*
- Invited to mentorship from Dr. Levin to extend "Structural Plasticity as Active Inference" research [3].
- Designed embodied agents that form a collective model and investigated robustness to damage & topological changes.

## Industry Research Engineer Experience

**Stealth Mode Startup**; Boston, MA *May 2025 - May 2026*
*Software Engineer (Research Engineer)* BrennenHill.com/stealth-startup
- Spearheaded the complete R&D for a novel AI system, taking the project from high-level goal to fully deployed system.
- Devised and implemented a custom hardware-aware algorithm that outperformed SOTA approaches by 100x accuracy.

**HRL Hughes Research Laboratories**; Malibu, CA *Summer 2024 & Summer 2023*
*Quantum Software Intern (Research Engineer)*
- **Quantum Compiler Optimization via Template Matching** BrennenHill.com/quantum-circuit
  - Translated novel research into a production-ready pattern matching algorithm within the Quilc quantum compiler.
  - Reduced quantum circuit depth by up to 37%, shortening execution time and mitigating qubit decoherence.
- **Low-Level Quantum Control Compiler** BrennenHill.com/quantum-compiler
  - Architected a multi-pass compiler in Common Lisp to generate optimized binary for a custom quantum control processor (QICK tProcessor ISA), maximizing performance and control flexibility for FPGA execution.
  - Implemented a custom assembly language to bridge high-level experimental logic with the hardware instruction set, enabling compiler optimizations, automatic resource allocation, and picosecond-level timing calculations.
- Awarded a return offer after each role in recognition of significant technical contributions and research impact.

## Representative Publications: First-Author, Peer-Reviewed, & Full-Length

1 **Brennen A. Hill**, Mant Koh En Wei, Thangavel Jishnuanandh. "Communicating Plans, Not Percepts: Scalable Multi-Agent Coordination with Embodied World Models ." *In proceedings of **NeurIPS 2025** Workshop on Scaling Environments for Agents; in NeurIPS 2025 Workshop on Embodied World Models for Decision Making; and in NeurIPS 2025 Workshop on Optimization for Machine Learning*. arxiv.org/abs/2508.02912

Developed a novel intention communication framework in MARL that achieved >96% success in a complex coordination task using a learned world model for latent trajectory planning and a self-attention mechanism to encode and share agent intentions, significantly outperforming emergent protocols.

2 **Brennen A. Hill**, Zhang Xinyu, Timothy Putra Prasetio. "The Geometry of Cortical Computation: Manifold Disentanglement and Predictive Dynamics in VCNet ." *In proceedings of **NeurIPS 2025** Workshop on Symmetry and Geometry in Neural Representations and in NeurIPS 2025 Workshop on Interpreting Cognition in Deep Learning Models*. arxiv.org/abs/2508.02995

Designed VCNet, a novel architecture emulating the primate visual cortex, achieving state-of-the-art accuracy on two vision benchmarks with over 10x greater parameter efficiency than standard models.

3 **Brennen A. Hill**. "Structural Plasticity as Active Inference: A Biologically-Inspired Architecture for Homeostatic Control." *In **National Science Foundation** (NSF) Workshop on Brain-Inspired Dynamics for Engineering Energy-Efficient Circuits and Artificial Intelligence*. arxiv.org/abs/2511.02241

One of two undergraduates selected to present alongside principal investigators. Developed a novel architecture that integrates synaptic and structural plasticity, demonstrating that agents can solve tasks by physically migrating processing units to minimize local prediction error, driven solely by active inference.

4 **Brennen A. Hill**. "Co-Evolving Complexity: An Adversarial Framework for Automatic MARL Curricula ." *In proceedings of **NeurIPS 2025** Workshop on Scaling Environments for Agents*. arxiv.org/abs/2509.03771

Developed a novel adversarial co-evolution framework to automatically generate a curriculum for multi-agent reinforcement learning that induces complex emergent strategies, increasing task performance by over 300%.

## Additional Publications: First-Author, Peer-Reviewed, & Full-Length

5 **Brennen A. Hill**. "The Physical Basis of Prediction: World Model Formation in Neural Organoids via an LLM-Generated Curriculum ." *In proceedings of **NeurIPS 2025** Workshop on Scaling Environments for Agents and in NeurIPS 2025 Workshop on Embodied World Models for Decision Making*. arxiv.org/abs/2509.04633

6 **Brennen A. Hill**. "Generative World Models of Tasks: LLM-Driven Hierarchical Scaffolding for Embodied Agents." *In **NeurIPS 2025** Workshop on Embodied World Models for Decision Making*. arxiv.org/abs/2509.04731

## Publications: In Review, First-Author, & Full-Length

7 **Brennen A. Hill**. "HEFT: A Coarse-to-Fine Hierarchy for Enhancing the Efficiency and Accuracy of Language Model Reasoning." *In review*. arxiv.org/abs/2509.09801

8 **Brennen A. Hill**, Surendra Parla, Venkata Abhijeeth Balabhadruni, Atharv Prajod Padmalayam, Sujay Chandra Shekara Sharma. "Breaking to Build: A Threat Model of Prompt-Based Attacks for Securing LLMs." *In review*. arxiv.org/abs/2509.04615

## Selected Research Projects

**Representation Fine-Tuning for Vision-Language Models** *Sep 2024 - Dec 2024*
*Lead Researcher* BrennenHill.com/vision-representations
- Investigated parameter-efficient Representation Fine-Tuning on a vision-language model for spatial reasoning.
- Co-authored a research paper achieving accuracy near LoRA (65.7% vs. 66.0%) with 10x fewer trainable parameters.

**Reinforcement Learning (RL) for Quadruped Roller Skating** *Sep 2024 - Dec 2024*
*Researcher* BrennenHill.com/rl-skating
- Co-authored a research paper on training a Unitree Go1 quadruped robot to skate using RL in the Isaac Gym simulator.
- Developed an RL policy resulting in emergent complex behaviors, including automatic gait switching across speeds.

## Honors and Awards

**Annenberg Fellowship** *May 2026*
*University of Southern California (USC)*
- Highly competitive fellowship awarded based on exceptional merit.
- "The faculty in your program believe you have the potential to make a significant contribution to your field."

**Viterbi School of Engineering & Graduate School Fellowship** *Feb 2026*
*University of Southern California (USC)*
- Greatly competitive fellowship awarded based on outstanding merit.
- Declined to accept Annenberg Fellowship

**UW-Madison Computer Science Department Interview & Feature** *Apr 2026*
*Featured Student Researcher* cs.wisc.edu/2026/04/08/brennen-hill-x26-is-building-ai-inspired-by-the-brain
- Highlighted in a dedicated feature article for founding the Wisconsin Neuromorphic Computing & NeuroAI Lab and advancing research on biologically-inspired artificial intelligence.

**RoboCup International Robotics Competition** *May 2024 & May 2025*
*1st Place (Shield 2024), 3rd Place (Standard Platform League 2025)*
- Won competition that is an international benchmark for multi-agent AI in adversarial environments, using NAO robots.
- Guided the team's technical strategy by conducting a comprehensive literature review on multi-robot soccer.
- Designed multi-agent RL coordination protocols for passing, applying concepts from my research [1].

**National University of Singapore School of Computing Showcase** *May 2025*
*Presenter* BrennenHill.com/ai-game
- Selected to present a self-developed 3D videogame to an audience of over 100 students and faculty.
- The game featured AI-driven agents with complex pathfinding and targetting algorithms in dynamic 3D environments.

**University of Wisconsin-Madison** *Sept 2022 - May 2026*
*Dean's List, Honors in Computer Science Program, Letters & Science Honors Program*

**NASA International Space Apps Challenge 2024** *Oct 2024*
*Honorable Mention* BrennenHill.com/exosky
- Received a global Honorable Mention, only awarded to 19 of 93,520 (0.02%) global participants.
- Won the Chicago hackathon and advanced to global finals; additionally awarded Best Presentation.
- Processed large-scale NASA/ESA astronomical data and implemented the 3D visualization & user interface.

**Hack Midwest 2024** *Oct 2024*
*Winner (awarded $2,500)* BrennenHill.com/badger-vision
- Won (of over 300 developers), awarded $2,500, and noted for Best Enterprise-Scale Business Solution.
- Built an AI tool for prosopagnosia (face blindness) by providing real-time audio for face and emotion identification.
- Implemented a computer vision pipeline using deep convolutional neural networks for a live low-level video stream.

**Agoura High School** *Aug 2018 - May 2022*
*Valedictorian of 600, 4.6/4.0 GPA, International Baccalaureate Diploma, State Golden State Seal Merit Award, State Seal of Biliteracy, 4-year Scholar Athlete*

## Professional Research Service

**Conference on Neural Information Processing Systems (NeurIPS)** *Dec 2026*
*Served as a peer reviewer upon nomination by the respective program committees:*
• Neurips 2025 Workshop on Symmetry and Geometry in Neural Representations (NeurReps)

**Conference on Language Modeling** *Oct 2026*
*Served as a peer reviewer upon nomination by the respective program committees:*
• COLM 2026 Workshop on Efficient Reasoning (ER)

**AAAI Conference on Artificial Intelligence** *Nov 2025*
*Served as a peer reviewer upon nomination by the respective program committees:*
• AAAI 2026 Workshop on Neuro for AI & AI for Neuro: Towards Multi-Modal Natural Intelligence (NeuroAI)

**Conference on Neural Information Processing Systems (NeurIPS)** *Sep 2025*
*Served as a peer reviewer upon nomination by the respective program committees:*
• Neurips 2025 Workshop on Scaling Environments for Agents (SEA)
• Neurips 2025 Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET)
• Neurips 2025 Workshop on Interpreting Cognition in Deep Learning Models (CogInterp)
• Neurips 2025 Workshop on Efficient Reasoning (ER)
• Neurips 2025 Workshop on Data on the Brain and Mind Findings (DBM)
• Neurips 2025 Workshop on Symmetry and Geometry in Neural Representations (NeurReps)

**Cortical Labs** *Oct 2024*
*Journal Club Host and speaker* BrennenHill.com/scale-free-cognition
• Presented a critical analysis on intelligence origins, scale-free cognition, and developmental bioelectricity.
• Facilitated a post-talk discussion with researchers on the future of synthetic biological intelligence.
• Invited to give future talks in recognition of the presentation's quality and the engaging discussion.

## Leadership & Professional Experience

**Madison Machine Learning**; *Co-Founder* *Sep 2025 - May 2026*
• Co-organized and grew a new community hub connecting machine learning researchers.
• Co-led weekly technical deep-dives and critical discussions on SOTA papers in machine learning.

**Badger Solar Racing**; *Head of AI* *Aug 2025 - May 2026*

**Badger Ballroom Dance Team**; *Vice President and Vice-Captain* *Dec 2023 - May 2026*

**AI Club**; *Executive Boardmember and Webmaster* *Mar 2024 - Mar 2025*

**Ballroom Association UW-Madison**; *Executive Boardmember and Webmaster* *Dec 2022 - Dec 2023*

**Slow Food UW**; *Volunteer Leadership* *Nov 2022*
• Mobilized and coordinated a team of 30 to prepare and distribute pay-what-you-can meals.

**Agoura High School Math Honors Society**; *Math Tutor* *Aug 2020 - May 2022*

**Agoura High School Science Honors Society**; *Science Tutor* *Aug 2020 - May 2022*

**Thunder Warrior Gaming**; Los Angeles, CA *Mar 2018 - Apr 2022*
*Lead Developer and Founder* BrennenHill.com/thunder-warrior
• Designed, built, self-published, and marketed the video game Thunder Warrior: Genesis, acheving net profit.
• Engineered custom game engine, multiplayer server, database, 3D models, game logic, and animations.

## Languages

**English**; *Native*

**Spanish**; *Fluent Written, Advanced Oral*